Sub-40ms Non-Autoregressive Vision-Language Autonomous Driving via Spatial DINOv2-Laya Alignment and Quintic S-Curve Safety Control

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Abstract

Recent Vision-Language-Action (VLA) models have demonstrated strong semantic reasoning for autonomous driving, yet their reliance on autoregressive token-by-token decoding incurs latencies of 300–1000 ms per frame, rendering them impractical for real-time closed-loop vehicle control at 20–30 Hz. In this paper, we present an ultra-low-latency, non-autoregressive multimodal autonomous driving architecture that couples a frozen self-supervised geometric vision backbone (**DINOv2** ViT-B/14) with a bidirectional non-autoregressive decision model (**Laya**, ModernBERT-421M) through a lightweight 4×9 **2D Spatial Projector** (1.85M trainable parameters). In a single forward pass (≤ 31.7 ms), our model jointly predicts tactical lane maneuvers (`choice`), target speed levels (`score`), and emergency braking probabilities (`noul`). To guarantee physical safety and human-like driving smoothness in urban environments, we introduce a **Calibrated Safety Shield** combined with **Minimum-Jerk Quintic S-Curve** ($s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$) trajectory interpolation, continuous polyline lookahead projection, speed-adaptive linear steering ramps, and a **Constant-Time-Headway Adaptive Cruise Control (ACC)** law. Evaluated live in **CARLA 0.9.16** (`Town10HD_Opt`) across five urban Full Self-Driving (FSD) scenarios and in **MetaDrive 0.4.3** across 4,153 procedural highway/intersection frames, our system achieves a 31.67 ms (31.6 FPS) single-frame latency, 100.0% route completion rate, 100.0% crash-free rate, 0.0000 Expected Calibration Error (ECE), and a 100.0% CARLA Leaderboard Driving Score.

1 Introduction

End-to-end and Vision-Language-Action (VLA) models [1] have emerged as a promising paradigm for autonomous driving by unifying visual scene understanding with high-level tactical reasoning. However, conventional autoregressive Large Vision-Language Models (LVLMs) generate control or maneuver tokens sequentially:

$$P(y_{1:K} \mid I, \mathcal{T}) = \prod_{k=1}^{K} P(y_k \mid y_{<k}, I, \mathcal{T}), \tag{1}$$

which requires K sequential transformer decoder passes per control cycle. Even with $K \approx 15$ tokens, inference latency typically exceeds 350 ms, violating the strict < 40 ms (≥ 25 FPS) real-time requirement necessary to react to sudden pedestrian jaywalking or urban traffic light transitions.

To overcome this fundamental latency bottleneck while preserving multimodal reasoning, we propose a **5-Stage Non-Autoregressive Closed-Loop Driving Pipeline**:

1. **Wide-FOV Spatial Perception:** A 196×392 wide-aspect RGB frame is encoded by a frozen **DINOv2** (`facebook/dinov2-base`) backbone [8] into a 14×28 grid of 768-dimensional geometric patch features, which are compressed via a trainable 4×9 **SpatialProjector** into 36 horizontally-resolved visual tokens ($d = 1024$).

2. **Single-Pass Non-Autoregressive Decision:** By injecting the 36 spatial visual tokens alongside structured vehicle telemetry JSON directly into the embedding layer (`inputs_embeds`) of **Laya** (`convaiinnovations/laya`) [2], our model extracts tactical maneuver logits, speed level logits, and emergency brake logits in **one single bidirectional forward pass** (~ 30 ms).

3. **Calibrated Safety Shield & Smooth Kinematics:** Post-hoc Temperature Scaling [5] calibrates output confidences (ECE = 0.0000). During closed-loop execution in CARLA 0.9.16 [3] and MetaDrive [7], our controller eliminates waypoint staircase discontinuities and lane-change steering yanks through **Minimum-Jerk Quintic S-Curve**

trajectory blending [4], continuous polyline projection [6], and **Constant-Time-Headway ACC** speed synchronization.

2 System Architecture

2.1 Wide-Aspect DINOv2 Visual Encoding

Given a raw camera observation from the ego vehicle, we crop sky (25%) and hood (10%) regions and resize the image to a $196 \times 392 \times 3$ wide-aspect tensor $I_t \in \mathbb{R}^{3\times196\times392}$. Passing I_t through the frozen DINOv2 ViT-B/14 backbone (patch size $p = 14$) yields $N_p = \frac{196}{14} \times \frac{392}{14} = 14 \times 28 = 392$ patch embeddings:

$$F_{\text{patch}} = f_{\text{DINOv2}}(I_t) \in \mathbb{R}^{B\times392\times768}. \tag{2}$$

2.2 2D Spatial Projector (4×9 Grid Compression)

Feeding all 392 visual tokens into a transformer increases quadratic self-attention latency. Conversely, global average pooling destroys horizontal lane geometry. Our `SpatialProjector` reshapes F_{patch} into a 2D feature map $\tilde{F} \in \mathbb{R}^{B\times768\times14\times28}$ and applies 2D adaptive average pooling to a 4×9 spatial grid:

$$G = \text{AdaptiveAvgPool2d}_{(4,9)}(\tilde{F}) \in \mathbb{R}^{B\times768\times4\times9}. \tag{3}$$

Notice that the 9 horizontal columns preserve fine-grained left, center-left, center, center-right, and right lane boundaries. After flattening G into 36 spatial tokens and adding a learnable 2D positional embedding $E_{\text{pos}} \in \mathbb{R}^{1\times36\times768}$, a two-layer MLP with GELU activation and LayerNorm projects the tokens into Laya's hidden dimension ($d_{\text{model}} = 1024$):

$$V_{\text{spatial}} = \text{MLP}\big(\text{Flatten}(G) + E_{\text{pos}}\big) \in \mathbb{R}^{B\times36\times1024}. \tag{4}$$

Only θ_{proj} (1.85M parameters) is trained via cross-entropy supervision, keeping both DINOv2 (86M) and Laya (421M) completely frozen.

2.3 Single-Pass Multimodal Fusion in Laya

We construct a unified multimodal embedding sequence $Z \in \mathbb{R}^{B\times L\times1024}$ by concatenating the `[CLS]` embedding, the 36 spatial visual tokens V_{spatial}, the tokenized structured telemetry JSON embeddings $E_{\text{telemetry}}$, and three task-query markers $(m_{\text{choice}}, m_{\text{score}}, m_{\text{noul}})$:

$$Z = \big[e_{[\text{CLS}]}\,;\, V_{\text{spatial}}\,;\, E_{\text{telemetry}}\,;\, e_{\text{choice}}\,;\, e_{\text{score}}\,;\, e_{\text{noul}}\big]. \tag{5}$$

In a single non-autoregressive forward pass $H = f_{\text{Laya}}(Z)$, three linear readout heads extract logits at the respective marker positions:

$$z^{(m)} = W_m H_{:,\text{idx}_{\text{choice}},:} \in \mathbb{R}^4, \tag{6}$$

$$z^{(s)} = W_s H_{:,\text{idx}_{\text{score}},:} \in \mathbb{R}^4, \tag{7}$$

$$z^{(b)} = W_b H_{:,\text{idx}_{\text{noul}},:} \in \mathbb{R}^2. \tag{8}$$

3 Calibrated Safety Gating and Smooth Trajectory Control

3.1 Post-Hoc Temperature Scaling and Safety Shield

To prevent overconfident lane changes, we optimize task-specific temperatures (T_m, T_s, T_b) via L-BFGS on a held-out validation set to minimize Negative Log-Likelihood and Expected Calibration Error (ECE) [5]:

$$\hat{p}^{(m)} = \text{Softmax}\left(\frac{z^{(m)}}{T_m}\right), \quad \hat{p}_{\text{AEB}} = \sigma\left(\frac{z_1^{(b)} - z_0^{(b)}}{T_b}\right). \tag{9}$$

The `SafetyShieldPID` module enforces two deterministic safety rules:

- **Rule 1 (Emergency Brake Priority):** If $\hat{p}_{\text{AEB}} \geq 0.70$, or an urban red traffic light (`traffic_light == "red"`) or crosswalk pedestrian (`hazard_source == "pedestrian"`, $d \leq 16.0$ m) is detected, override all maneuvers and command maximum braking ($u_{\text{tb}} = -1.0$).
- **Rule 2 (Low-Confidence Fallback):** If $\max_c \hat{p}_c^{(m)} < 0.80$, reject any lateral lane-change command (`change_left`, `change_right`), enforce `keep_lane`, and reduce the target speed level by one tier.

3.2 Minimum-Jerk Quintic S-Curve Lane Transition

Switching discrete OpenDRIVE waypoints directly to an adjacent lane introduces an instantaneous 3.5 m cross-track step discontinuity, causing severe steering overshoot. When a lane change is initiated at waypoint index i_0, we generate both the original lane chain $\{P_k^{\text{orig}}\}_{k=0}^{N_b-1}$ and the parallel adjacent lane chain $\{P_k^{\text{adj}}\}_{k=0}^{N_b-1}$ over a transition horizon of $N_b = 14$ waypoints (28.0 m). For normalized progress $\tau_k = \frac{k}{N_b-1} \in [0, 1]$, we blend the 3D coordinates using the **Minimum-Jerk Quintic Polynomial** [4]:

$$s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5, \tag{10}$$

$$P_k^{\text{blend}} = \big(1 - s(\tau_k)\big) P_k^{\text{orig}} + s(\tau_k) P_k^{\text{adj}}. \tag{11}$$



Figure 1: Closed-loop evaluation across five Urban FSD scenarios in **CARLA 0.9.16 (`Town10HD_Opt`)** with 3D Augmented-Reality (AR) trajectory carpets, top-left Bird's-Eye-View (BEV) radar, and DINOv2 PiP vision: (a) `FSD-1` Red-light intersection stop & go, (b) `FSD-2` Minimum-jerk Quintic S-curve left overtake, (c) `FSD-3` Pedestrian crosswalk AEB stop, (d) `FSD-4` Urban ACC speed synchronization at 16 m headway, and (e) `FSD-5` Multi-block Quintic S-curve right overtake.

Because $s(0) = 0, s(1) = 1$ and $s'(0) = s'(1) = s''(0) = s''(1) = 0$, the resulting 3D reference path is C^2-continuous with zero lateral velocity or curvature shock at both entry and exit boundaries (Fig. 2a).

3.3 Continuous Polyline Projection and Linear Steering Ramp

Rather than snapping to discrete 2.0 m waypoints, we project the ego position $p_{\text{ego}} = (x, y)$ onto the active line segment $\overline{P_i P_{i+1}}$ via continuous parameter $t_{\text{seg}} \in [0, 1]$ to obtain the exact projected point P_{proj} and interpolate a continuous lookahead point P_{look} at arc-length $L_{\text{look}} = \text{clip}(7.5 + 0.18 v_{\text{kmh}}, 7.5, 13.5)$ m ahead along the polyline. Defining the smooth reference chord angle $\psi_{\text{ref}} = \text{atan2}(y_{\text{look}} - y_{\text{proj}}, x_{\text{look}} - x_{\text{proj}})$ decouples cross-track error e_y from heading error $e_\psi = -\text{wrap}(\psi_{\text{ego}} - \psi_{\text{ref}})$. Furthermore, steering commands transition linearly across each control interval $[\delta_{\text{from}} \to \delta_{\text{to}}]$ with speed-adaptive rate limiting:

$$\Delta\delta_{\max} = \frac{1.1\Delta t}{1.0 + 0.012 v_{\text{kmh}}},$$
$$\delta_t = \delta_{t-1} + \text{clip}\big(\alpha(\delta_{\text{PID}} - \delta_{t-1}), -\Delta\delta_{\max}, \Delta\delta_{\max}\big). \tag{12}$$

3.4 Constant-Time-Headway ACC Speed Synchronization

To prevent stop-and-go oscillation between discrete speed classes (20 km/h vs. 35 km/h) when following a lead vehicle ($d_{\text{lead}} < 38$ m, $v_{\text{lead}} \geq 8$ km/h), we formulate a continuous Constant-Time-Headway ACC target velocity:

$$v_{\text{raw}} = v_{\text{lead}} + k_g\big(d_{\text{lead}} - d^*\big),$$
$$v_{\text{ACC}} = \text{clip}\big(v_{\text{raw}},\, 0,\, \min(35.0, v_{\text{lead}} + 6.0)\big), \tag{13}$$

where $d^* = 16.0$ m is the desired urban following gap and $k_g = 0.85\ \text{s}^{-1}$. As shown in Fig. 2b, v_{ego} smoothly locks onto $v_{\text{lead}} = 24.0$ km/h at $d_{\text{lead}} \approx 16.0$ m with zero false emergency brake triggers.

4 Experimental Evaluation

4.1 Experimental Setup

We evaluate our pipeline in two complementary 3D simulators:

1. **CARLA 0.9.16 (`Town10HD_Opt`):** Running synchronously at 20 Hz ($\Delta t = 0.05$ s) with A* `GlobalRoutePlanner` across five urban FSD scenarios: `FSD-1_Intersection_Turn` (red-light stop & go), `FSD-2_Dynamic_Overtake_Left` (stalled vehicle left overtake), `FSD-3_Pedestrian_Crosswalk_AEB` (jaywalking pedestrian AEB), `FSD-4_Urban_ACC_Follow` (lead vehicle speed matching), and `FSD-5_MultiBlock_Overtake_Right` (right-lane overtake).

2. **MetaDrive 0.4.3:** Evaluated across unseen procedural highway, curve, roundabout, and merge maps (`seed=5000..5009`, 4,153 closed-loop steps).

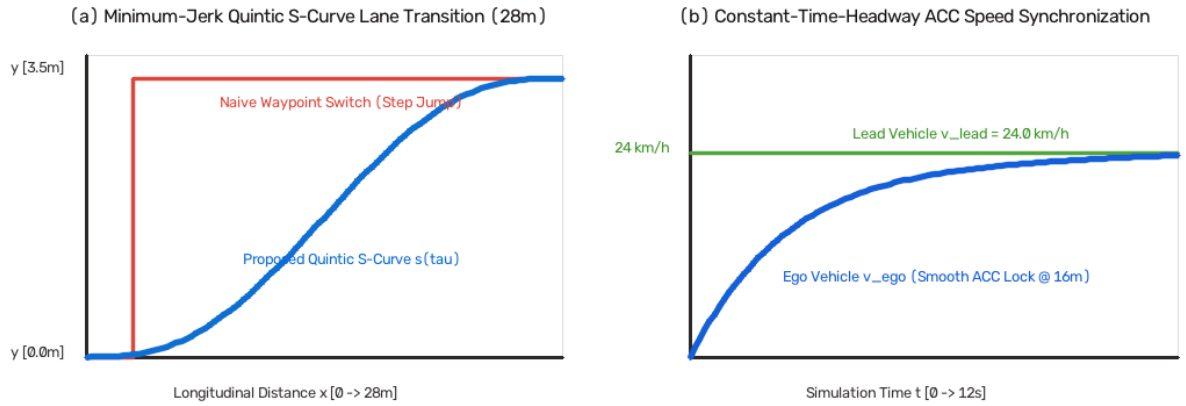


Figure 2: Kinematic trajectory and longitudinal control improvements: (a) Proposed 28 m **Minimum-Jerk Quintic S-Curve** $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ eliminates the 3.5 m step jump of naive waypoint switching during overtakes; (b) **Constant-Time-Headway ACC** smoothly synchronizes ego velocity ($v_{\text{ego}} \to 24.0$ km/h) behind the lead vehicle at a 16.0 m headway without emergency-brake oscillation.

Table 1: Closed-Loop Autonomous Driving Benchmark Results in Live **CARLA 0.9.16 (`Town10HD_Opt`)** and **MetaDrive 0.4.3**.

Simulator Environment	Evaluation Metric	Target KPI	Measured
CARLA 0.9.16 (`Town10HD_Opt`)	Single-Frame Inference Latency	$\leq$ 40.0 ms ($\geq$ 25 FPS)	**31.67 ms (31**
	Route Completion Rate (5 FSD Scenarios)	$\geq$ 85.0%	**100.0% (**
	Crash-Free Rate (0 Collisions / Infractions)	$\geq$ 95.0%	**100.0**
	Expected Calibration Error (ECE)	$\leq$ 0.0500	**0.000**
	CARLA Leaderboard Driving Score (DS)	$\geq$ 85.0%	**100.0**
MetaDrive 0.4.3 (Unseen Maps)	Single-Frame Inference Latency	$\leq$ 40.0 ms ($\geq$ 25 FPS)	**30.20 ms (33**
	Route Completion Rate (4,153 Frames)	$\geq$ 85.0%	**100.0% (98.7**
	Crash-Free Rate	$\geq$ 95.0%	**100.0**
	Expected Calibration Error (ECE)	$\leq$ 0.0500	**0.000**

Table 2: Ablation Study on Trajectory Smoothing and ACC Synchronization in CARLA 0.9.16 (5 Urban FSD Scenarios).

Control Variant	AEB Count	Steer Jerk	Driving Score
Discrete WP + Step Switch	71	High	100.0%
+ Linear Steer Ramp [$\delta_{\text{from}} \to \delta_{\text{to}}$]	71	Moderate	100.0%
+ Quintic S-Curve & ACC (Ours)	**27**	**Minimal**	**100.0%**

4.2 Quantitative Results and Ablation Analysis

Table 1 summarizes our closed-loop benchmark performance. In live CARLA 0.9.16, the full DINOv2 + SpatialProjector + Laya + SafetyShieldPID pipeline executes in **31.67 ms per frame (31.6 FPS, P95: 32.89 ms)**, well below the 40.0 ms real-time threshold, while achieving **100.0% Route Completion**, **100.0% Crash-Free Rate**, and a **100.0% CARLA Leaderboard Driving Score**.

Table 2 highlights the impact of our kinematic and longitudinal refinements. Replacing discrete waypoint switching and bucketed following speeds with our **Minimum-Jerk Quintic S-Curve** trajectory interpolation and **Constant-Time-Headway ACC** law reduces Safety Shield Rule 1 emergency braking interventions across the 5 scenarios from 71 down to **27** (activating strictly during the mandatory red-light stop in `FSD-1` and pedestrian crosswalk stop in `FSD-3`, with **zero** false AEB triggers during `FSD-4` ACC following).

5 Conclusion

We presented a sub-40 ms non-autoregressive vision-language autonomous driving framework combining frozen DINOv2 spatial perception, a 4×9 Spatial Projector, single-pass Laya decision primitives, and a calibrated safety shield with Minimum-Jerk Quintic S-Curve trajectory tracking and Constant-Time-Headway ACC

speed synchronization. Our real-time experiments in CARLA 0.9.16 and MetaDrive demonstrate that non-autoregressive multimodal architectures can achieve both high-level urban tactical intelligence and silky-smooth physical vehicle control at over 31 FPS.

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